

Continuous-Wave Backreflection Measurement of Polarization Mode Dispersion

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Abstract—Experimental results of continuous-wave polarization mode dispersion (PMD) measurements are presented. The proposed technique is based on the backscattered signal given by the Fresnel reflection at the fiber far-end. Experimental statistical distribution of the differential group delay of the round-trip is reported. It presents a Rayleigh distribution whose mean value is simply related to the fiber mean differential group delay. Finally, the statistical distribution of the round-trip dispersion vector is found experimentally and numerically. The proposed method can be easily used for one-end measurement of PMD in installed optical cables.

Index Terms—Optical fiber communication, optical fiber measurements, optical polarization, polarization mode dispersion, optical fiber reflectometry.

I. INTRODUCTION

UP to now, polarization mode dispersion (PMD) has been measured by means of several techniques in the time domain and in the frequency domain [1]. All these techniques allow measurement of the differential group delay (DGD), although only Jones-Matrix Eigenanalysis (JME) and Poincaré Sphere method (PS) retrieve information on PMD statistical properties. The main drawback is that they require both fiber ends during measurements, and as a consequence, field tests on installed optical cables are time consuming and not simple. This is the main reason why a different approach based on polarization sensitive backscattering technique has been investigated.

Although polarization optical time domain reflectometry (POTDR) has been introduced about twenty years ago (see, e.g., [2]), its usefulness in the complete PMD characterization of long fibers has been shown only recently. In fact, with POTDR it is possible to measure, at the same time, the mean value of the beat length and the DGD [3], [4] for fibers in the *short-length regime* as well as in the *long-length regime* [5]. However, POTDR usually requires a rather complicated experimental setup, which nowadays is not yet easy to use in field tests.

In this letter, a new simple PMD measurement scheme is presented, based on the backreflected signal due only to the Fresnel reflection at the fiber far-end. Such a solution requires a continuous wave tunable source and a commercially available

polarization analyzer, and it enables one to measure PMD in fibers up to 40 km long, using only one end of the link. Experimental results support the theoretical prediction that the DGD of the round-trip $\Delta\tau_B$ presents a Rayleigh distribution and its mean value is $\pi/2$ times the mean DGD of the fiber, $\Delta\tau$.

II. THEORETICAL BACKGROUND

When a continuous optical wave is injected into a fiber, the total backscattered power is given by the sum of two contributions: the Rayleigh scattering, and the Fresnel reflection due to the fiber far-end face. The Rayleigh backscattered power can be expressed as follows [6]:

$$P_S \simeq P_0 \frac{S\alpha_R}{2\alpha} (1 - \exp(-2\alpha L)),$$

where symbols have the following meaning (values are typical for step-index single-mode fibers at 1550 nm): P_0 is the input power [W], L is the fiber length [m], $S \simeq 1.5 \cdot 10^{-3}$ represent the recapture factor, $\alpha_R \simeq 2.76 \cdot 10^{-5} \text{ m}^{-1}$ is the Rayleigh scattering loss coefficient, and $\alpha \simeq 5 \cdot 10^{-5} \text{ m}^{-1}$ is the attenuation coefficient.

Similarly, the contribution due to the power reflected by the fiber end face is given by

$$P_F = P_0 \rho_F \exp(-2\alpha L)$$

where $\rho_F \simeq 0.035$ is the reflection coefficient of the glass-air interface.

In order to perform a PMD measurement based on the power reflected by the fiber far-end face, it should be assumed that P_F is the backscattered power dominant component. Setting a minimum acceptable value of the ratio P_F/P_S , the maximum fiber length is consequently fixed; as an example, when $P_F/P_S > 5 \text{ dB}$, then $L < 40 \text{ km}$. Therefore, from now on it will be assumed that the total backscattered power is due to P_F only. The maximum fiber length can be further increased placing a perfect reflector at the fiber far-end (i.e., $\rho_F \rightarrow 1$).

Let $\hat{s}_0$ be the unitary Stokes vector of the input field, and $\mathbf{R}$ the Muller matrix of a single-mode fiber; the SOP of the output field can be expressed as $\hat{s} = \mathbf{R}\hat{s}_0$. The evolution of $\hat{s}$ as a function of frequency is well known [7]:

$$\frac{d\hat{s}}{d\omega} = \frac{d\mathbf{R}}{d\omega} \mathbf{R}^{-1} \hat{s} = \bar{\Omega} \times \hat{s} \quad (1)$$

where $\mathbf{R}^{-1}$ represents the inverse matrix of $\mathbf{R}$. Vector $\bar{\Omega} = (\Omega_1, \Omega_2, \Omega_3)$ is the polarization dispersion vector, and $\Delta\tau = |\bar{\Omega}|$.

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A similar result can be obtained for the SOP of the back-reflected field $\hat{s}_B$. Indeed it results [3]: $\hat{s}_B = \mathbf{R}_B \hat{s}_0 = \mathbf{M} \mathbf{R}^T \mathbf{M} \hat{s}_0$, where $\mathbf{M} = \text{diag}(1, 1, -1)$ is a diagonal matrix and $\mathbf{R}^T$ is the transpose of $\mathbf{R}$. The analogy can be pushed further, as it is easy to prove that the evolution of $\hat{s}_B$ as a function of frequency is described by [3]

$$\frac{d\hat{s}_B}{d\omega} = \frac{d\mathbf{R}_B}{d\omega} \mathbf{R}_B^{-1} \hat{s}_B = \bar{\Omega}_B \times \hat{s}_B \quad (2)$$

where $\bar{\Omega}_B$ may be written as

$$\bar{\Omega}_B = 2\mathbf{M} \mathbf{R}^T \begin{pmatrix} \Omega_1 \\ \Omega_2 \\ 0 \end{pmatrix}. \quad (3)$$

Since $\Delta\tau_B = |\bar{\Omega}_B|$, and $\mathbf{M}$ and $\mathbf{R}$ are orthogonal matrices, it is easy to find [3]

$$\Delta\tau_B = 2\sqrt{\Omega_1^2 + \Omega_2^2}.$$

It is well known that, when the fiber is in the *long-length regime*, $\Omega_i, i = 1, 2, 3$, are Gaussian distributed random variables, statistically independent of each other, with zero mean and the same standard deviation [7]. Because of this, $\Delta\tau$ is Maxwellian while $\Delta\tau_B$ results are Rayleigh distributed and its probability density function (pdf) reads as follows [3]:

$$f(\Delta\tau_B) = \frac{2\Delta\tau_B}{\pi\langle\Delta\tau\rangle^2} \exp\left(-\frac{\Delta\tau_B^2}{\pi\langle\Delta\tau\rangle^2}\right) \quad (4)$$

where $\langle\Delta\tau\rangle$ is the mean value of $\Delta\tau$. Finally, the following relationship holds:

$$\langle\Delta\tau_B\rangle = \frac{\pi}{2}\langle\Delta\tau\rangle. \quad (5)$$

It can be noticed that the motion equation of the round-trip, (2), formally coincides with the motion equation of forward propagation, (1). Consequently $\bar{\Omega}_B$ can be measured applying the same techniques used for forward measurements. Even if (5) is independent of the particular technique, we decided to use JME because it can measure not only $\langle\Delta\tau_B\rangle$, but also each element of $\bar{\Omega}_b$ [8].

III. EXPERIMENTAL RESULTS

The schematic diagram of the experimental setup is shown in Fig. 1. The optical source is an external cavity laser (output power 0 dBm), tuned between 1480 and 1570 nm, in steps of 1 nm. The SOP of the source radiation is modified by a polarization controller, which is connected to a low-loss polarization-insensitive directional coupler (port 1), so that the light is launched into the fiber under test throughout port 3 of the coupler. This port needs to be terminated with an angled connector (FC-APC) in order to reduce the near-end reflection to less than 60 dB. Finally a polarization analyzer is connected to port 2 of the coupler (for round-trip measurements) or alternatively to the fiber far-end (for forward measurements).

A first set of measurements was performed on an ensemble of single-mode step-index fibers, spliced together to obtain cascades with different lengths, ranging from 2 km up to 40 km; $\Delta\tau$ and $\Delta\tau_B$ were measured by means of the JME

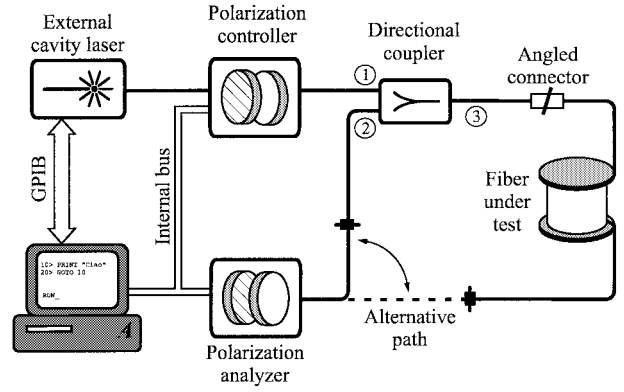


Fig. 1. Schematic diagram of the experimental setup.

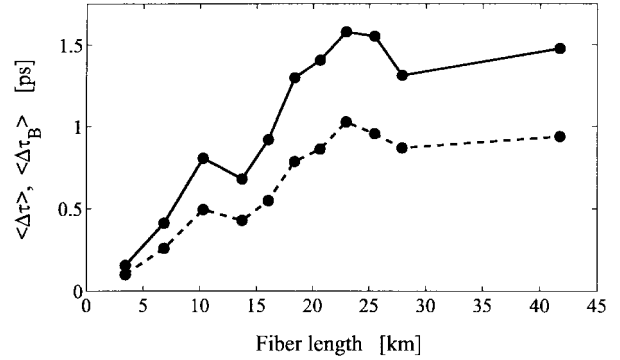


Fig. 2. Evolution of $\langle\Delta\tau_B\rangle$ (continuous line) and $\langle\Delta\tau\rangle$ (dashed line) as a function of the fiber length.

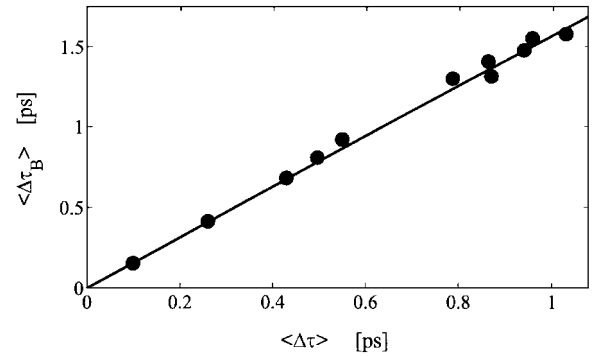


Fig. 3. Plot of $\langle\Delta\tau_B\rangle$ versus $\langle\Delta\tau\rangle$; continuous line is obtained by means of (5).

method. Results are shown in Fig. 2, where values of $\langle\Delta\tau\rangle$ and $\langle\Delta\tau_B\rangle$ are plotted as function of the cascade length. Averaging was performed over wavelength.

The same data are reported in Fig. 3; black dots are the experimental points and they fit very well the continuous line that is the theoretical function predicted by (5).

In a second experiment the statistical properties of $\bar{\Omega}_B$ have been found by means of the JME method. The fiber (single-mode, dispersion-shifted, 6 km long) was inside a temperature-controlled chamber, where the temperature was forced to cycle up and down between 3 and 54 °C, in steps of 3 °C. At each step the temperature was stabilized for 30 min.; afterwards the measure was performed, scanning wavelength

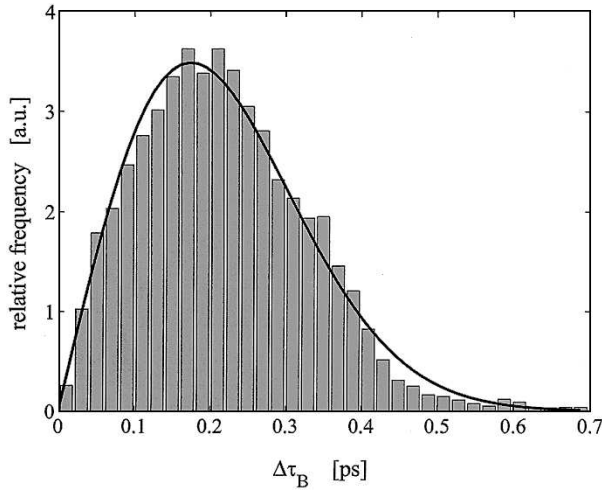


Fig. 4. The pdf of $\Delta\tau_B$: vertical bars are experimental data, continuous line refers to (4).

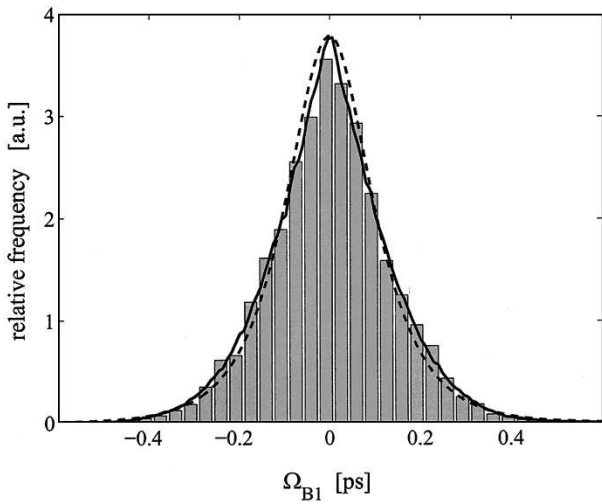


Fig. 5. The pdf of Ω_{B1} : vertical bars are experimental data, continuous line was numerically obtained, dashed line refers to (6).

from 1480 to 1570 nm, in steps of 1 nm. Measurements were repeated 80 times, in order to obtain an ensemble of 7200 experimental points.

The experimental histogram of $\Delta\tau_B$ (i.e., the modulus of $\bar{\Omega}_B$) is reported in Fig. 4. The continuous line shows how well the Rayleigh distribution fits the experimental data.

Fig. 5 shows the experimental histogram of Ω_{B1} , the first component of $\bar{\Omega}_B$; the other two components present the same shape and they are not reported for the sake of brevity.

The continuous line was numerically obtained simulating an ensemble of 300 000 fibers, such that the mean value of forward DGD was the same as that obtained from measured data (i.e., $\langle\Delta\tau\rangle = 0.13$ ps). This was achieved by means of the same wave-plate model adopted in [5], using 100 randomly oriented plates per fiber. The DGD of each plate was set to 14 fs, so that the overall mean DGD could coincide with the experimental value. Again, there is a very good agreement between experimental and numerical results. The analytical expressions of the Ω_{Bi} pdf's ($i = 1, 2, 3$) are quite difficult to find, since the statistical properties of matrix $\mathbf{R}$ involved in (3) are unknown. However, a good approximation is given by the following pdf (shown in Fig. 5 with dotted line):

$$f(\Omega_{Bi}) = \frac{1}{2\sigma} \operatorname{sech}\left(\frac{\pi\Omega_i}{2\sigma}\right) \quad (6)$$

where σ is the standard deviation of the experimental data.

IV. CONCLUSION

Experimental results have been presented about a PMD measurement scheme, based on the analysis of the Fresnel reflection originated by the fiber far-end face. The DGD of the round-trip presents a Rayleigh distribution, whose mean value is $\pi/2$ the mean value of the fiber DGD. Experimental data fit very well with theoretical and numerical predictions, confirming the usefulness and the capability of this technique for field tests on installed links, using only one fiber end.

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